

Constraining GLM trends from MMEs using observations

Introduction

Method

We have a set of m climate simulations, x^j with $j = 1, \dots, m$. Each x^j is a timeseries of a climate variable available in the historical period and the future. We also have a set of observations z which are only available in the historical period.

The observations are distributed as a glm with some location parameter ν_z :

$$z_t | \nu(t) \sim \text{GLM}(\nu_z(t)), \quad t = t_1, \dots, t_{n_h}, \quad (1)$$

$$g(\nu_z(t)) = a_1 + a_2 t. \quad (2)$$

The climate simulations have mean ν_{x^j} :

$$x_t^j | \nu_{x^j}(t) \sim \text{GLM}(\nu_{x^j}(t)), \quad t = t_1, \dots, t_{n_f}. \quad (3)$$

$$g(\nu_{x^j}(t)) = b_1^j + b_2^j t, \quad j = 1, \dots, m. \quad (4)$$

We can approximate these to:

$$\hat{b}^j \sim \text{MVN}(b^j, \text{Var}(\hat{b}^j)), \quad j = 1, \dots, m. \quad (5)$$

$$\hat{a} \sim \text{MVN}(a, \text{Var}(\hat{a})). \quad (6)$$

Where $\hat{a}$ and $\text{Var}(\hat{a})$ are the MLE and Covariance matrix of the MLE for regression parameters $a = (a_1, a_2)^T$.

Then assuming a Gaussian latent model:

$$b^j | \mu, \Sigma \sim \text{MVN}(\mu, \Sigma), \quad j = 1, \dots, m. \quad (7)$$

To relate a to b^j use:

$$b^0 | \mu, \Sigma \sim \text{MVN}(\mu, \Sigma). \quad (8)$$

$$a = b^0 + w, \quad (9)$$

where:

$$w | \Gamma \sim \text{MVN}(0, \Gamma), \quad (10)$$

and Γ is defined in terms of Σ and $\delta = (\delta_1, \delta_2)^T$, the similarity parameters:

$$\Gamma_{ll} = \Sigma_{ll} \frac{\delta_l - 1}{\delta_l}, \quad l = 1, 2, \quad (11)$$

$$\Gamma_{1,2} = \Gamma_{2,1} = 0.$$

When $\delta = 1$ $\Gamma = 0$ and as $\delta \rightarrow 0$ $\Gamma \rightarrow \infty$.

We can then either fix δ or have a prior distribution over it:

$$\delta_l \sim \text{Beta}(\theta_l^{(1)}, \theta_l^{(1)}), \quad l = 1, 2. \quad (12)$$

If we want priors on μ and Σ :

$$\mu \sim N(\mu_\mu, \sigma_\mu), \quad (13)$$

$$\Omega \sim LKJ(\eta), \quad (14)$$

$$\tau \sim \text{half-Cauchy}(0, \gamma), \quad (15)$$

$$\Sigma = \text{diag}(\tau)\Omega\text{diag}(\tau). \quad (16)$$

Simulated data

Lets set up the time covariate, t in years, which we rescale to t_sc .

```
start_date <- 1980
end_hist <- 2014
end_date <- 2100
nh <- end_hist - start_date + 1
nf <- end_date - start_date + 1
t <- seq(start_date, end_date, length.out = nf)
t_sc <- (t - start_date)/(nf - 1)
```

Lets generate some simulated regression parameters (intercepts and slopes) where we know the true Σ , μ and w .

```
m = 20 # number of simulations
set.seed(1234)

Sigma_true <- cbind(c(.02, -.01), c(-.01, .2))
mu_true <- cbind(2, 0)
L_true <- t(chol(Sigma_true))

# The true simulation regression parameters
b_true <- matrix(rep(mu_true, m), nrow = 2) + L_true %*% matrix(rnorm(m * 2), nrow = 2)
w_true <- c(0.5, -0.3)

# The true real world regression parameters
a_true <- as.numeric(mu_true) + L_true %*% rnorm(2) + w_true

print("mu is:")

## [1] "mu is:"
print(mu_true)

##      [,1] [,2]
## [1,]    2    0
print("Sigma is:")

## [1] "Sigma is:"
print(Sigma_true)
```

```
##      [,1] [,2]
## [1,]  0.02 -0.01
## [2,] -0.01  0.20
```

Now lets simulate the data as a Poisson distributed GLM.

```
B <- cbind(rep(1, nf), t_sc)

X <- matrix(rpois(n = m * nf, exp(as.numeric(B %*% b_true))), ncol = m)

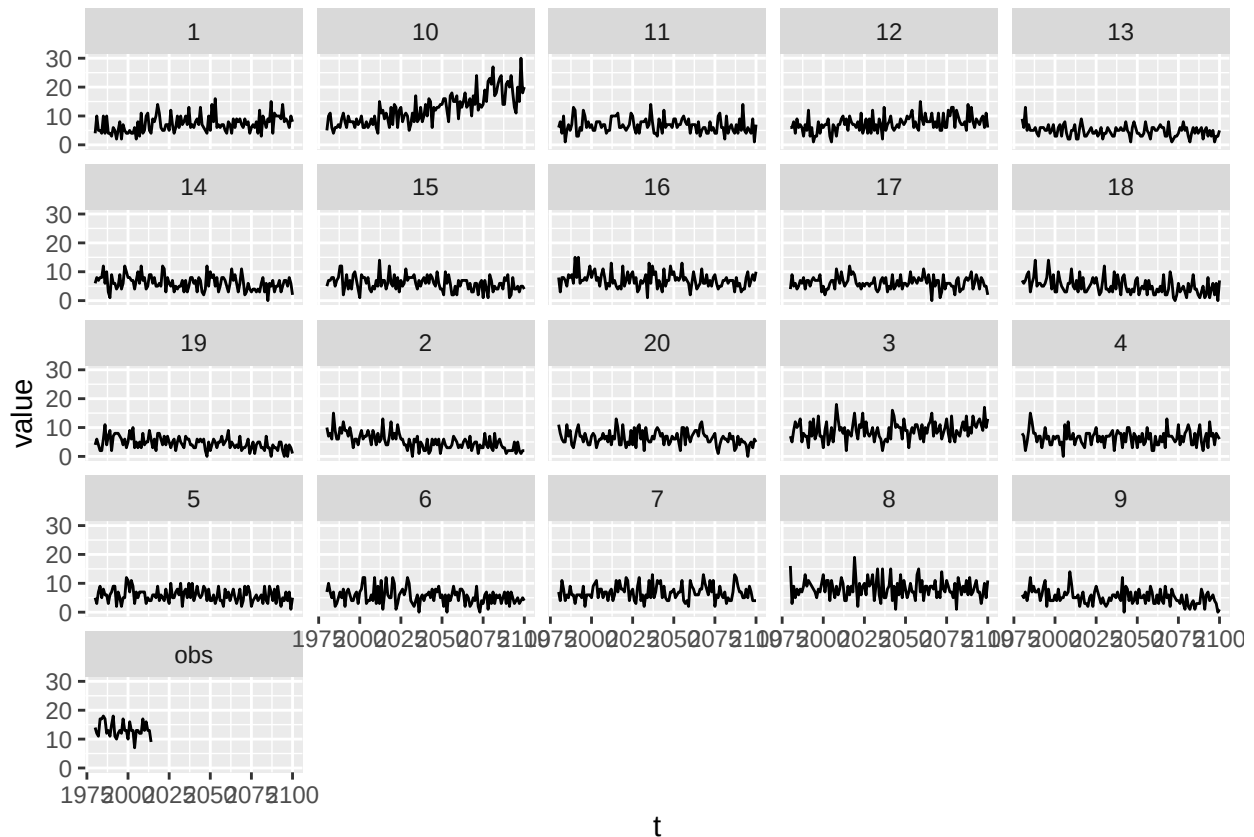
z <- rpois(nh, exp(as.numeric(B[1:nh,] %*% a_true)))
```

And put into a data frame:

```
data_df <- data.frame(X) %>%
  mutate(t_sc = t_sc, t = t) %>%
  pivot_longer(-c(t_sc, t), names_to="model", names_prefix="X") %>%
  bind_rows(
    data.frame(value = z, t_sc = t_sc[1:nh], t = t[1:nh], model = "obs")
  )
```

Lets plot this data

```
data_df %>%
  ggplot(aes(x = t, y = value)) +
  geom_line() +
  facet_wrap(~model)
```



Fitting the model

We first need to fit a GLM to all the simulation time series and the observed time series.

```
fit_list <- list()
for(j in 1:m){
  fit_list[[j]] <- glm(
    value ~ t_sc,
    poisson(link = "log"),
    filter(data_df, model == j)
  )
}
a_fit <- glm(value ~ t_sc, poisson(link = "log"), filter(data_df, model == "obs"))
```

Then we need to get the MLEs for the regression parameters and the lower Cholesky decomposition of their covariance matrices.

```
b_hat <- matrix(ncol = 2, nrow = m)
J <- array(0, dim = c(m, 2, 2))
for(j in 1:m){
  b_hat[j,] <- fit_list[[j]]$coef
  J[j,,] <- t(chol(vcov(fit_list[[j]])))
}
a_hat <- a_fit$coef
J_a <- t(chol(vcov(a_fit)))
```

Then we create a `data_list` to pass to `cmdstan`. In this case if `len_hyper_param` is 0 and the length of the `prior_hyper_param` vector is 0, the stan code assumes infinite uniform priors for μ and Σ .

Lets choose the $\delta_1 = \delta_2 = 1$ case.

```
prior_hyper_param = numeric() # an empty vector will result in no priors
# prior_hyper_param = c(
#   mu_mu_1 = 0,
#   mu_mu_2 = 0,
#   mu_sd_1 = 10,
#   mu_sd_2 = 5,
#   eta = 2,
#   gam = 2.5
# ) # an length 6 vector of prior hyper parameters

data_list <- list(
  len_a = 2,
  m = m,
  b_hat = b_hat,
  a_hat = a_hat,
  J = J,
  J_a = J_a,
  len_hyper_param = length(prior_hyper_param),
  hyper_param = prior_hyper_param,
  delta_hyper_param = c(1,1,1,1), # Prior parameters for delta ~ Beta() prior
  delta = c(1,1) # Fixed values of delta
)

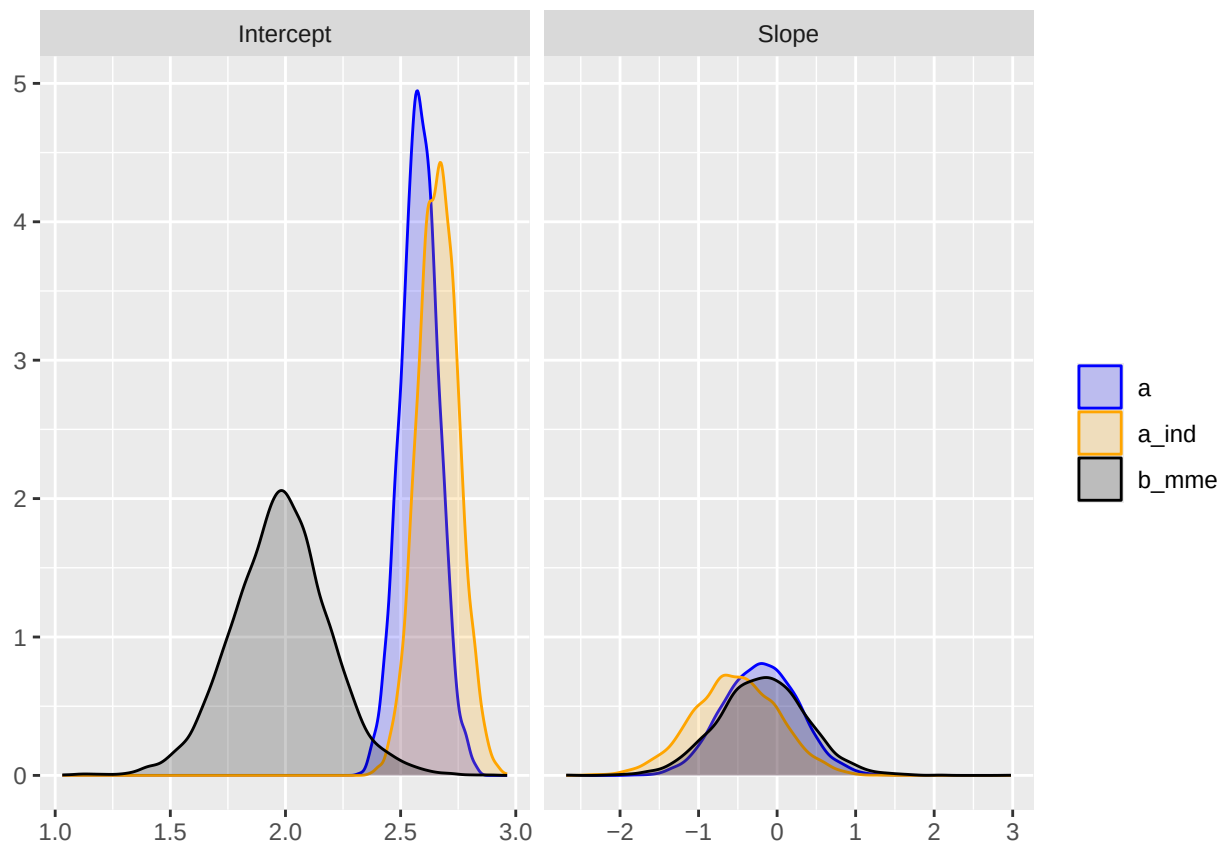
# This stan model fits the hierarchical model for a fixed delta
mod <- cmdstan_model("stan/glm_constr_mme_obs_fixed_delta.stan")
```

```
fit_fixed_delta <- mod$sample(
  data = data_list,
  seed = 1234,
  iter_warmup = 1000,
  iter_sampling = 1000,
  chains = 4,
  parallel_chains = 4,
  refresh = 0,
  show_exceptions = FALSE
)
```

```
## Running MCMC with 4 parallel chains...
##
## Chain 2 finished in 4.3 seconds.
## Chain 3 finished in 4.3 seconds.
## Chain 1 finished in 4.4 seconds.
## Chain 4 finished in 4.4 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 4.4 seconds.
## Total execution time: 4.7 seconds.
```

Lets look at the intercept and slope predicted for the MME of climate simulations ($b_{mme} \sim MVN(\mu, \Sigma)$), the observations independent of the MME ($a_{ind} \sim MVN(\hat{a}, Var(\hat{a}))$) and the posterior from the model (a):

```
## Warning: Dropping 'draws_df' class as required metadata was removed.
```



Let now consider a number of different δ values. Note the model cannot accept a value of $\delta = 0$.

```
deltas = c(0.001,seq(.25,1,.25))
delta_mat <- cbind(rep(deltas, 5),rep(deltas, each = 5))
delta_mat

##      [,1] [,2]
## [1,] 0.001 0.001
## [2,] 0.250 0.001
## [3,] 0.500 0.001
## [4,] 0.750 0.001
## [5,] 1.000 0.001
## [6,] 0.001 0.250
## [7,] 0.250 0.250
## [8,] 0.500 0.250
## [9,] 0.750 0.250
## [10,] 1.000 0.250
## [11,] 0.001 0.500
## [12,] 0.250 0.500
## [13,] 0.500 0.500
## [14,] 0.750 0.500
## [15,] 1.000 0.500
## [16,] 0.001 0.750
## [17,] 0.250 0.750
## [18,] 0.500 0.750
## [19,] 0.750 0.750
## [20,] 1.000 0.750
## [21,] 0.001 1.000
## [22,] 0.250 1.000
## [23,] 0.500 1.000
## [24,] 0.750 1.000
## [25,] 1.000 1.000

prior_hyper_param = numeric()           # an empty vector will result in no priors
prior_hyper_param = c(
  mu_mu_1 = 0,
  mu_mu_2 = 0,
  mu_sd_1 = 10,
  mu_sd_2 = 5,
  eta = 2,
  gam = 2.5
)   # an length 6 vector of prior hyper parameters

data_list_mult_delta <- list(
  len_a = 2,
  m = m,
  b_hat = b_hat,
  a_hat = a_hat,
  J = J,
  J_a = J_a,
  len_hyper_param = length(prior_hyper_param),
  hyper_param = prior_hyper_param,
  n_delta = nrow(delta_mat),
  delta = delta_mat
)   # Fixed values of delta
```

```
# This stan model fits the hierarchical model for a set of fixed deltas.
mod <- cmdstan_model("stan/glm_constr_mme_obs_fixed_delta_vector.stan")
```

```
fit_fixed_delta_mult_delta <- mod$sample(
  data = data_list_mult_delta,
  seed = 1234,
  iter_warmup = 1000,
  iter_sampling = 1000,
  chains = 4,
  parallel_chains = 4,
  refresh = 0,
  show_exceptions = FALSE
)
```

```
## Running MCMC with 4 parallel chains...
##
## Chain 3 finished in 13.8 seconds.
## Chain 4 finished in 14.1 seconds.
## Chain 2 finished in 14.4 seconds.
## Chain 1 finished in 15.2 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 14.4 seconds.
## Total execution time: 15.3 seconds.
```

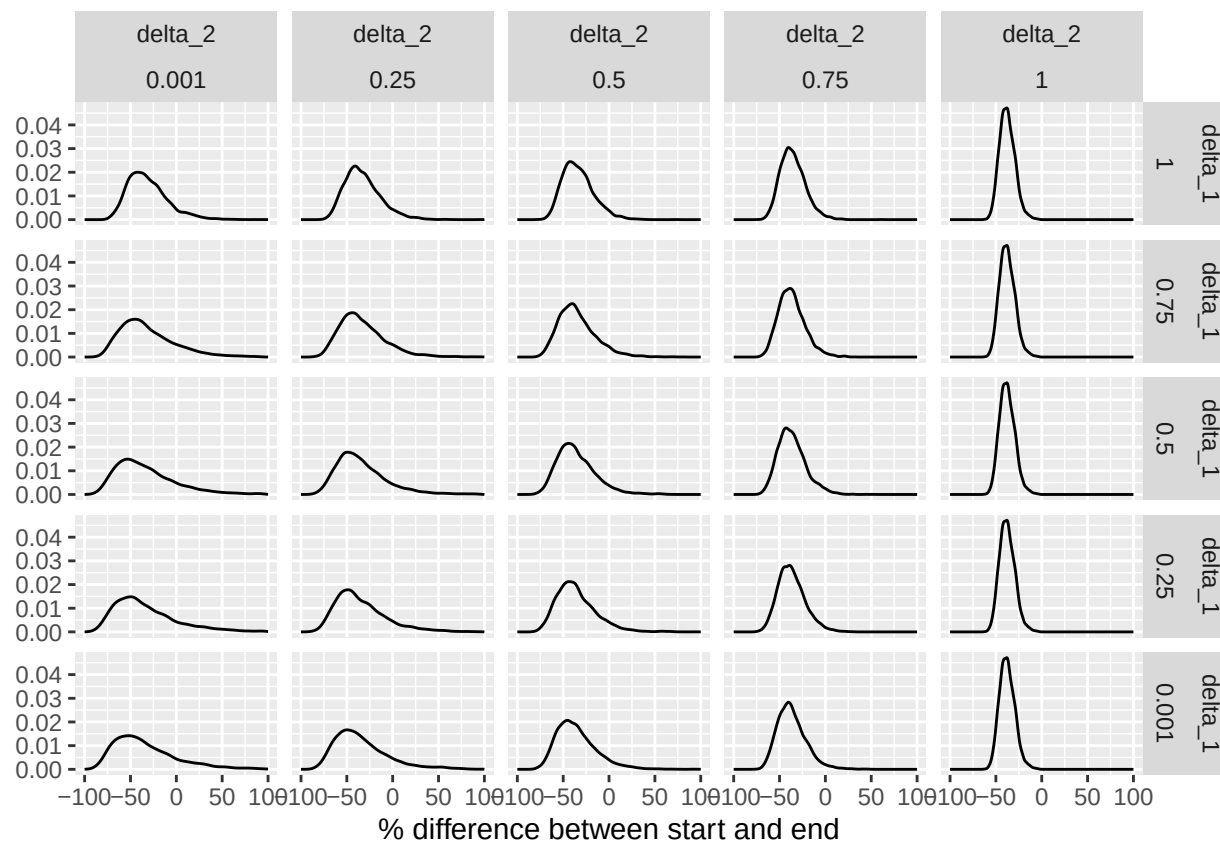
Lets use precentage change from the start of the time period to the end of the time period ($t_{sc} = 0$, $t = 1980$ to $t_{sc} = 1980$, $t = 2100$) as an interesting example:

$$100\% \frac{E(z|t = 2100) - E(z|t = 1980)}{E(z|t = 1980)} = 100\% \frac{\exp(a_1 + a_2) - \exp(a_1)}{\exp(a_1)} \quad (17)$$

$$= 100\%(\exp(a_2) - 1) \quad (18)$$

Plotting for the different choices of δ

```
## Warning: Dropping 'draws_df' class as required metadata was removed.
## Warning: Removed 166 rows containing non-finite outside the scale range
## (`stat_density()`).
```



Finally we can sample from the posterior given a prior on δ !

Here we do a uniform prior $\delta_l \sim \text{Unif}(0, 1), l = 1, 2$:

```
mod <- cmdstan_model("stan/glm_constr_mme_obs.stan")
```

```
fit_prior_on_delta <- mod$sample(
  data = data_list,
  seed = 1234,
  iter_warmup = 1000,
  iter_sampling = 1000,
  chains = 4,
  parallel_chains = 4,
  refresh = 0,
  show_exceptions = FALSE
)
```

```
## Running MCMC with 4 parallel chains...
##
## Chain 1 finished in 6.3 seconds.
## Chain 3 finished in 6.2 seconds.
## Chain 2 finished in 6.4 seconds.
## Chain 4 finished in 6.6 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 6.4 seconds.
## Total execution time: 6.8 seconds.
```

```
## Warning: Dropping 'draws_df' class as required metadata was removed.
```

